GLITTERS Property Valuation Intelligence Engine (v1.1.0)

Phase 1 Specification: Evidence Integrity, Multi-Tenant Model Isolation, Causal Conformal Calibration & Scalable Kenyan Market Architecture

Principal Quantitative Research, Computational Economics & Machine Learning Engineering Specification

Version 1.1.0-KE | Package: `glitters_valuation_engine` | Target Currency: KES (Metric m^2)

Abstract

This specification defines the mathematical foundations, econometric estimators, spatial stochastic processes, machine learning ensembles, uncertainty quantification protocols, and production integration architecture for **GLITTERS Valuation Engine v1.1.0** (`glitters_valuation_engine`), incorporating all **Phase 1: Evidence Integrity & Statistical Correctness** audit remediations. Designed for Kenya's property markets across all 47 counties, the engine unifies seven valuation paradigms while enforcing five critical mathematical and engineering invariants: (1) strict segregation of capital sale evidence from rental cash-flow observations; (2) immutable SHA-256 model training provenance that blocks synthetic data from production training by default and propagates synthetic provenance into every inference output; (3) model-level multi-tenant scope binding (`PUBLIC_REGISTRY_ONLY`, `SINGLE_TENANT_PRIVATE`, `AUTHORIZED_POOLED`) preventing cross-tenant weight leakage; (4) exact finite-sample order-statistic conformal calibration $k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ requiring $n_{\text{cal}} \geq \lceil (1 - \alpha)/\alpha \rceil$ coupled with locally weighted KNN conformal scaling; and (5) global chronological sorting prior to locality MAD outlier screening and strictly causal historical reference filtering ($t_{\text{comp}} \leq t_{\text{val}}$) with zero temporal lookahead leakage.

Statutory & Non-Certification Compliance Invariant: All outputs generated by `glitters_valuation_engine` are quantitative decision-support estimates and do **not** constitute a statutory valuation report certified by a Registered and Practising Valuer under the Valuers Act (Cap 532, Laws of Kenya) or the Institution of Surveyors of Kenya (ISK) Valuation Standards (2021). By default, `PropertyRecord.title_status` initializes to `UNVERIFIED_MISSING` and requires an explicit `TitleVerificationMetadata` audit record from official Ministry of Lands (Ardhisasa) searches before a title is treated as verified.

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1 Phase 1 Technical Audit Remediations (CRITICAL 1–5)

Table 1 summarizes the engineering and mathematical remediations implemented in `glitters_valuation_engine` v1.1.0 in response to the Technical Audit.

Table 1: Summary of Phase 1 Technical Audit Remediations in `glitters_valuation_engine` v1.1.0

Audit Finding	Root Cause Identified in v1.0.0	Mathematical & Code Remediation in v1.1.0	Verification Test
CRITICAL 1: Rental observations contaminating sale valuations	`PriceEvidenceType` included rental contracts/quotes in the same pool consumed by Comparable Sales, Hedonic, Spatial GP, and GBDT.	Partitioned evidence into `CAPITAL_SALE_EVIDENCE_TYPES` and `RENTAL_EVIDENCE_TYPES`. Sale models reject rental records; `IncomeValuationEngine` consumes rental records with explicit `RentalPeriodUnit` conversion (`MONTHLY`, `ANNUAL`, `PER_SQM_MONTHLY`).	`test_critical_01_*`
CRITICAL 2: Synthetic training data mistaken for market evidence	`fit()` accepted synthetic data by default; synthetic generator used `PUBLIC_REGISTRY`; output flag only checked subject and comparables.	Default `allow_synthetic_training=False` raises `PermissionError` on synthetic input. Synthetic generator uses `SYNTHETIC_SANDBOX`. `ModelProvenance` records `trained_on_synthetic_data` and SHA-256 dataset hash, propagating into `is_synthetic_evidence`.	`test_critical_02_*`
CRITICAL 3: Tenant isolation missing on trained model weights	Comparable pool filtered by tenant, but Hedonic/Spatial/ML weights trained on Tenant A could be queried by Tenant B.	Added `ModelTrainingScope` (`PUBLIC_REGISTRY_ONLY`, `SINGLE_TENANT_PRIVATE`, `AUTHORIZED_POOLED`) and `authorized_tenant_ids`. Unauthorized tenant queries return `REJECTED_UNAUTHORIZED_TENANT_SCOPE`. Enforced `tx.tenant_org_id == prop.tenant_org_id`.	`test_critical_03_*`
CRITICAL 4: Invalid small-sample conformal quantile & global-only multiplier	Capping quantile level at 1.0 when $n_{\text{cal}} < 9$ violated finite-sample coverage; only global multiplier implemented.	Implemented exact order statistic $k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ requiring $n_{\text{cal}} \geq \lceil (1 - \alpha)/\alpha \rceil = 9$ for 90% coverage. If $n_{\text{cal}} < 9$, sets `is_conformal_calibrated=False`. Added locally weighted KNN conformal multiplier $\hat{q}_{1-\alpha}(\mathbf{z}_*)$.	`test_critical_04_*`
CRITICAL 5: Locality-grouped train/cal split & temporal lookahead leakage	Locality MAD filter reordered records by locality before slicing `cal_txs`; calibration used full reference pool.	Deduplication and locality MAD filter record rejected IDs and re-sort survivors in global chronological order ($t_1 \leq \cdots \leq t_N$). Calibration evaluates each record at t_{cal} using only comparables with $t_{\text{comp}} \leq t_{\text{cal}}$.	`test_critical_05_*`
HIGH PRIORITY: Packaging, Sparse GP, Config Priors & Title Default	Missing `pyproject.toml`; dense $\mathcal{O}(N^3)$ GP; hard-coded priors; `VERIFIED_CLEAN` default title status.	Added `pyproject.toml`, `setup.py`, `README.md`; Nyström Inducing-Point Sparse GP ($\mathcal{O}(NM^2)$, $M = 64$); `ValuationMarketConfig.calibrate_from_transactions()`; default `UNVERIFIED_MISSING` title status + `TitleVerificationMetadata`.	`test_high_priority_*`

2 Core Valuation Problem & Kenyan Market Formulation

Let $\mathcal{P}$ denote the space of real property parcels and built units across Kenya's 47 counties. Each subject property $i \in \mathcal{P}$ is represented at valuation timestamp $t \in \mathbb{R}$ by the canonical feature tuple:

$$\mathcal{X}_i(t) = \big(\mathbf{s}_i, c_i, \mathbf{x}_{\text{struct},i}, \mathbf{x}_{\text{loc},i}, \mathbf{x}_{\text{zone},i}, \mathbf{x}_{\text{tenure},i}, \mathbf{x}_{\text{inc},i}, \mathbf{x}_{\text{macro}}(t)\big), \tag{1}$$

where:

- $\mathbf{s}_i = (\phi_i, \lambda_i) \in [-4.75°, 5.05°] \times [33.90°, 41.95°]$ is the WGS84 latitude-longitude coordinate pair mapped to county, sub-county, and neighbourhood locality $L(i)$.
- $c_i \in \mathcal{C}$ is the property category across ten mandatory classes:

$$\mathcal{C} = \{\text{Residential}, \text{Commercial}, \text{Industrial}, \text{Agricultural}, \text{Vacant Land}, \\ \text{Apartment/Condo}, \text{Rental}, \text{Development Land}, \text{Investment}, \text{Distressed}\}.$$

- $\mathbf{x}_{\text{struct},i} = (A_{\text{land},i}, A_{\text{bldg},i}, B_{\text{bed},i}, B_{\text{bath},i}, \text{Age}_i, Q_{\text{cond},i}, Q_{\text{const},i})$ captures metric land area $A_{\text{land},i}$ (m^2), gross building floor area $A_{\text{bldg},i}$ (m^2), bedrooms, bathrooms, physical age Age_i (years), condition score $Q_{\text{cond},i} \in [0,1]$, and construction quality $Q_{\text{const},i} \in [0,1]$.
- $\mathbf{x}_{\text{loc},i} = (I_{\text{access},i}, I_{\text{infra},i}, I_{\text{util},i}, \text{Tier}_i, d_{\text{CBD}}(\mathbf{s}_i))$ captures accessibility, paved road/drainage infrastructure, utilities (piped water, mains sewer, 3-phase electricity, fibre), neighbourhood tier $\text{Tier}_i \in \{1,2,3,4\}$ (aligned with Kenya Bankers Association Housing Price Index (KBA-HPI) regional clustering), and great-circle distance to the nearest urban economic hub.
- $\mathbf{x}_{\text{zone},i} = (Z_i, \text{GC}_{\max,i}, \text{FAR}_{\max,i}, H_{\max,i})$ defines permitted zoning category Z_i, maximum ground coverage ratio $\text{GC}_{\max,i} \in (0,1]$, maximum Plot Ratio / Floor Area Ratio $\text{FAR}_{\max,i} > 0$, and maximum storeys $H_{\max,i}$ under the Physical and Land Use Planning Act, No. 13 of 2019 (PLUPA) and county development control policies such as the Nairobi City County Development Control Policy (Sessional Paper No. 1 of 2023).
- $\mathbf{x}_{\text{tenure},i} = (\tau_i, T_{\text{lease},i}, \Omega_{\text{title},i}, \mathcal{M}_{\text{title},i})$ specifies tenure type $\tau_i \in \{\text{Freehold}, \text{Leasehold}, \text{Sectional Title}, \text{Customa}$ unexpired leasehold term $T_{\text{lease},i}$ (years), title verification state $\Omega_{\text{title},i}$ (defaulting to `UNVERIFIED_MISSING`), and optional audit metadata $\mathcal{M}_{\text{title},i}$ (`registry_reference`, `verification_source`, `verified_timestamp_i` `evidence_document_hash`).

2.1 Strict Separation of Capital Sale Evidence vs. Rental Evidence

The evidence space $\mathcal{E}$ is partitioned into two disjoint subsets:

$$\mathcal{E}_{\text{sale}} = \{\texttt{VERIFIED_STAMP_DUTY_SALE}, \texttt{VERIFIED_CONTRACT_SALE}, \texttt{VALUER_HISTORICAL_RECORD}, \texttt{ASKING_LISTIN} \tag{2}$$

$$\mathcal{E}_{\text{rent}} = \{\texttt{RENTAL_LEASE_CONTRACT}, \texttt{RENTAL_ASKING_QUOTE}\}, \qquad \mathcal{E}_{\text{sale}} \cap \mathcal{E}_{\text{rent}} = \emptyset. \tag{3}$$

Observations with $E_j \in \mathcal{E}_{\text{sale}}$ enter the capital-value training and comparable-sales pipelines, with asking prices adjusted by configurable negotiation haircut $\kappa_{\text{ask}} \in$ `ValuationMarketConfig` (empirically estimable via `ValuationMarketConfig.calibrate_from_transactions()`). Observations with $E_j \in \mathcal{E}_{\text{rent}}$ carry an explicit `RentalPeriodUnit` $\in \{\texttt{MONTHLY}, \texttt{ANNUAL}, \texttt{PER_SQM_MONTHLY}\}$ and are routed exclusively to `IncomeValuationEngine.estimate_market_rent_from_comparables()`.

2.2 Structural Leasehold Tenure Relativity

We define the actuarial leasehold relativity factor $f_{\text{lease}}(\tau_i, T_{\text{lease},i}) \in (0, 1]$ relative to perpetual freehold at real discount rate r_L (default 0.09), with a statutory renewal friction deduction when $T_{\text{lease},i} < 25$ years:

$$f_{\text{lease}}(\tau_i, T_{\text{lease},i}) = \begin{cases} 1.00, & \text{if } \tau_i = \text{Freehold}, \\ (1 - (1 + r_L)^{-T_{\text{lease},i}}) - 0.08 \max\Big(0, \dfrac{25 - T_{\text{lease},i}}{25}\Big), & \text{if } \tau_i \in \{\text{Leasehold}, \text{Sectional Title}\}. \end{cases} \tag{4}$$

To guarantee strict economic monotonicity across all valuation paradigms (verified across 60 randomized trials in `test_06_sixty_trial_property_based_fuzzing`), $f_{\text{lease}}(\tau_i, T_{\text{lease},i})$ is applied structurally as a multiplicative relativity factor across Hedonic, Spatial GP, GBDT, Income, Cost, and Residual Land models rather than as an unconstrained collinear regression covariate.

3 Mathematical Formulations of the Seven Valuation Approaches

3.1 Approach A: Semi-Log Regularized Hedonic Model with Duan's Smearing

Fitted strictly on capital sales $\mathcal{D}_{\text{sale}}$, the hierarchical semi-log hedonic model with Empirical Bayes locality random intercepts $\alpha_{L(i)}$ and standardized structural/locational covariates $\tilde{\boldsymbol{\phi}}(\mathcal{X}_i) \in \mathbb{R}^{22}$ is:

$$\ln V_i = \beta_0 + \alpha_{L(i)} + \tilde{\boldsymbol{\phi}}(\mathcal{X}_i)^\top \boldsymbol{\beta} + \varepsilon_i, \qquad \hat{V}_{\text{hedonic},i} = \exp(\hat{\beta}_0 + \hat{\alpha}_{L(i)} + \tilde{\boldsymbol{\phi}}(\mathcal{X}_i)^\top \hat{\boldsymbol{\beta}}) \cdot \hat{S}_{\text{Duan}} \cdot f_{\text{lease}}(\tau_i, T_i), \tag{5}$$

where $\hat{S}_{\text{Duan}} = (\sum_{j=1}^{N} w_j \exp(\hat{e}_j)) / \sum_{j=1}^{N} w_j$ corrects Jensen's inequality retransformation bias under non-normal residuals.

3.2 Approach B: Causal Comparable Sales Intelligence Engine

For subject property i valued at timestamp t_0, candidate j is admitted if and only if $E_j \in \mathcal{E}_{\text{sale}}$ and $t_j \le t_0$ (strictly preventing future lookahead leakage). The composite kernel weight is:

$$K(i, j) = K_{\text{spatial}}(d_{ij}) \cdot K_{\text{temp}}(t_0 - t_j) \cdot K_{\text{struct}}(\mathbf{z}_i, \mathbf{z}_j) \cdot w_{\text{rel}}(E_j) \cdot \exp\big(-\gamma \max(0, \text{GAR}_{ij} - 0.25)\big). \tag{6}$$

From the top K comparables with normalized weights $\tilde{w}_{ij}$, the engine computes $\hat{V}_{\text{comp},i} = \sum_{j=1}^{K} \tilde{w}_{ij} P_{j \to i}^{\text{adj}}$, Effective Sample Size $N_{\text{eff}} = (\sum_j K_{ij})^2 / \sum_j K_{ij}^2$, and Comparable Quality Score $\text{CQS}_i \in [0, 1]$.

3.3 Approach C: Scalable Inducing-Point Sparse Spatial Gaussian Process

To eliminate the $\mathcal{O}(N^3)$ computational bottleneck and $\mathcal{O}(N^2)$ memory footprint of exact Kriging on large GLITTERS transaction histories, `SpatialGPValuationModel` implements the **Deterministic Training Conditional (DTC) / Nyström Inducing-Point Sparse Gaussian Process** over $M \le 64$ spatially stratified landmark knots $\mathbf{Z} = \{\mathbf{u}_1, \ldots, \mathbf{u}_M\} \subset \mathbb{R}^2$. Approximating the Matérn($\nu = 3/2$) covariance matrix as $\mathbf{K}_{NN} \approx \mathbf{K}_{NM} \mathbf{K}_{MM}^{-1} \mathbf{K}_{MN} + \sigma_n^2 \mathbf{I}_N$, the Woodbury matrix identity reduces inversion to the $M \times M$ system:

$$\boldsymbol{\Sigma}_M = \mathbf{K}_{MM} + \frac{1}{\sigma_n^2} \mathbf{K}_{MN} \mathbf{K}_{NM} \in \mathbb{R}^{M \times M}, \qquad \boldsymbol{\alpha}_M = \boldsymbol{\Sigma}_M^{-1} \left(\frac{1}{\sigma_n^2} \mathbf{K}_{MN} \mathbf{e}_{\text{trend}} \right). \tag{7}$$

At target location $\mathbf{s}_*$, the predictive mean and posterior variance are computed in $\mathcal{O}(M)$ time:

$$\mu_{\text{GP}}(\mathbf{s}_*) = \mathbf{x}(\mathbf{s}_*)^\top \hat{\boldsymbol{\beta}}_{\text{trend}} + \mathbf{k}_{*M}^\top \boldsymbol{\alpha}_M, \qquad \sigma_{\text{GP}}^2(\mathbf{s}_*) = (\sigma_f^2 + \sigma_n^2) - \mathbf{k}_{*M}^\top (\mathbf{K}_{MM}^{-1} - \boldsymbol{\Sigma}_M^{-1}) \mathbf{k}_{M*}. \tag{8}$$

In our empirical scalability benchmark ($N = 2{,}500$ transactions, $M = 64$ inducing points), offline fitting completes in **66.86 ms** and single-property online inference completes in **0.289 ms**.

3.4 Approaches D, E, F & G: GBDT/RF, Rental-Integrated Income DCF, Cost DRC & PLUPA 2019 Residual HBU

- **Second-Order GBDT + Bootstrap RF (`machine_learning`)**: Fits $M_{\text{tree}} = 35$ Newton-step boosted trees on $\mathcal{D}_{\text{sale}}$ alongside $B = 12$ bootstrap Random Forest trees to quantify non-parametric epistemic tree variance.
- **Income Capitalization & 10-Year DCF (`income_valuation`)**: Estimates market rent from nearby $\mathcal{E}_{\text{rent}}$ comparables ($t_j \le t_0$) when subject rent is absent, blends 75% contractual rent with 25% market rental comparables when both exist, and averages Direct Capitalization (NOI_1/R_0) with a 10-year DCF model benchmarked to the Central Bank of Kenya (CBK) risk-free rate.
- **Depreciated Replacement Cost (`cost_valuation`)**: Combines urban-gravity baseline land value with JBC/IQSK reproduction cost new less Ross-Heidecke physical deterioration, functional obsolescence, and external obsolescence.
- **Residual Land & HBU (`land_valuation`)**: Maximizes discounted residual land value across legally permissible uses under PLUPA 2019 Ground Coverage and Plot Ratio limits.

4 Exact Finite-Sample & Locally Adaptive Conformal Calibration

Let $\mathcal{D}_{\text{cal}} = \{(\mathcal{X}_j, V_j)\}_{j=1}^{n_{\text{cal}}}$ denote the chronologically held-out calibration cohort where $t_{\text{train}} \le t_{\text{cal},1} \le \cdots \le t_{\text{cal},n_{\text{cal}}}$. Each calibration observation j is evaluated using only historical comparables available at $t_{\text{cal},j}$, yielding normalized non-conformity score $s_j = |V_j - \hat{V}_{\text{ens},j}|/\sigma_{\text{total},j}$.

Theorem 4.1 (Exact Finite-Sample Conformal Order Statistic & Minimum Sample Condition)**.** *For target confidence level $1-\alpha \in (0,1)$, exchangeability implies $\mathbb{P}(s_{n_{cal}+1} \le s_{(k)}) = \frac{k}{n_{cal}+1} \ge 1-\alpha$ if and only if the order-statistic index k satisfies:*

$$k = \lceil (n_{cal}+1)(1-\alpha) \rceil \le n_{cal} \quad \Longleftrightarrow \quad n_{cal} \ge \left\lceil \frac{1-\alpha}{\alpha} \right\rceil. \tag{9}$$

For $1-\alpha = 0.90$ ($\alpha = 0.10$), the minimum valid calibration sample size is $n_{cal,\min} = 9$. Whenever $n_{cal} < 9$, a 90% finite-sample empirical order statistic does not exist; `UncertaintyAndConfidenceEngine` *therefore sets* `is_conformal_calibrated = False`*, emits an explicit* `UNCALIBRATED_PREDICTION_INTERVAL` *flag, and applies a conservative fallback multiplier without claiming conformal calibration.*

When $n_{\text{cal}} \ge n_{\text{cal,min}}$ and standardized calibration features $\{\mathbf{z}_j\}_{j=1}^{n_{\text{cal}}}$ are present, the engine computes a locally adaptive conformal multiplier $\hat{q}_{1-\alpha}(\mathbf{z}_*)$ around target $\mathbf{z}_*$ using Gaussian kernel weights $w_j(\mathbf{z}_*) = \exp(-\|\mathbf{z}_* - \mathbf{z}_j\|_2^2/(2Dh_z^2))$:

$$\hat{q}_{1-\alpha}(\mathbf{z}_*) = \max\Big(0.85\, s_{(k)},\ 0.70\, s_{(k)} + 0.30\, \hat{Q}^{\text{local}}_{1-\alpha}(\{(s_j, w_j(\mathbf{z}_*))\})\Big). \tag{10}$$

5 Empirical Benchmark Results, Coverage Audit & Scalability

Table 2 reports out-of-sample benchmark performance on the 360-record synthetic Kenyan multi-county dataset (`synthetic_kenya_property_dataset.csv`, explicitly labelled with `SYNTHETIC_SANDBOX` provenance) after applying all Phase 1 remediations.

Impact of Fixing Calibration Leakage (CRITICAL 4 & 5): After fixing the locality-grouping split bug and enforcing causal historical calibration ($t_{\text{comp}} \le t_{\text{cal}}$) with exact order-statistic conformal quantiles, the out-of-sample 90% prediction interval coverage (PICP_{90}) on the Rolling Temporal Benchmark improved from 79.22% (Calibration Error 10.78% in v1.0.0) to **89.61% (Calibration Error 0.39% against the 90.0% target in v1.1.0)**, while R^2 improved from 0.9387 to 0.9598 and MAE dropped to KES 12.52M.

Table 2: Phase 1 Remediated Out-of-Sample Benchmark Results on Synthetic Kenyan Dataset ($N = 360$ Sales + 51 Rentals)

Model / Valuation Approach	MAE (KES)	RMSE (KES)	MedAE (KES)	MAPE (%)	R^2	$PICP_{90}$ (%)	Cal. Err (%)	Pinball Loss (KES)
Panel A: Strictly Causal Rolling Origin Temporal Holdout (Future 25% Verified Sales)								
Baseline: Tier Median Price/m^2	34,855,905	63,927,085	13,646,098	83.84	0.5952	40.26	49.74	9,319,963
Spatial GP Matérn(3/2) Sparse Kriging	24,830,714	51,588,449	8,661,697	37.51	0.7364	45.45	44.55	5,795,239
Regularized Semi-Log Hedonic (Duan)	23,971,970	44,719,749	10,227,316	41.27	0.8019	42.86	47.14	4,422,217
Comparable Sales Kernel Engine	21,547,411	40,379,574	8,971,627	35.12	0.8385	59.74	30.26	3,958,424
Second-Order GBDT + Bootstrap RF	18,286,768	32,487,802	10,246,655	35.32	0.8955	51.95	38.05	3,519,998
GLITTERS v1.1.0 Full Ensemble	**12,515,363**	**20,142,976**	**7,675,291**	**31.95**	**0.9598**	**89.61**	**0.39**	**1,702,204**
Panel B: Spatial Neighbourhood Block Holdout (Unseen Localities: Ruaka, Milimani Kisumu, Syokimau)								
Comparable Sales (Zero Local Comps)	45,305,233	93,184,785	11,649,336	106.69	0.0199	44.74	45.26	13,695,185
Baseline: Tier Median Price/m^2	41,695,722	81,241,601	11,718,694	63.11	0.2550	30.26	59.74	14,376,303
Gradient Boosted Trees + RF	16,397,141	31,154,181	9,173,309	33.53	0.8905	53.95	36.05	3,375,834
Regularized Semi-Log Hedonic (Duan)	15,084,417	25,183,998	7,462,731	28.97	0.9284	59.21	30.79	2,266,641
Spatial GP Matérn(3/2) Sparse Kriging	15,008,422	26,193,991	7,897,099	29.59	0.9226	57.89	32.11	2,264,627
GLITTERS v1.1.0 Full Ensemble	**11,544,719**	**18,889,276**	**6,661,583**	**33.49**	**0.9597**	**100.00**	**10.00**	**1,768,426**

6 Recommended GLITTERS Integration Architecture & Package Layout

The package is installable via standard PEP 517/518 tooling (`pyproject.toml` and `setup.py`) and includes the `integration_adapter` module implementing the application-service boundary recommended in Section 6 of the Technical Audit:

```
glitters_package_root/
|-- pyproject.toml
|-- setup.py
|-- README.md
|-- GLITTERS_Valuation_Engine_Specification.tex
|-- GLITTERS_Valuation_Engine_Specification.pdf
`-- glitters_valuation_engine/
    |-- __init__.py
    |-- contracts.py                # Frozen contracts, ModelProvenance,
        TitleVerificationMetadata
    |-- config.py                   # ValuationMarketConfig + empirical calibration
        routine
    |-- data_validation/            # Chronological MAD filter, sale/rental split,
        tenant check
    |-- feature_engineering/        # 22-D spatial/structural/PLUPA pipeline
    |-- comparable_analysis/        # Causal capital-sale kernel matching, N_eff & CQS
    |-- hedonic_models/             # ElasticNet semi-log regression + Duan smearing +
        EB offsets
    |-- spatial_models/             # Vectorized Moran's I + Inducing-Point Sparse GP (M
        <=64)
    |-- machine_learning/           # Second-order GBDT + Bootstrap RF epistemic tracker
    |-- income_valuation/           # Rental comparable estimator + Direct Cap + 10-Yr
        DCF
    |-- cost_valuation/             # JBC/IQSK Depreciated Replacement Cost (Ross-
        Heidecke)
    |-- land_valuation/             # PLUPA 2019 Residual Land Value & HBU optimizer
    |-- ensemble/                   # Tenant-scoped, provenance-tracked regime meta-
        learner
    |-- uncertainty/                # Exact order-statistic & locally weighted conformal
         engine
    |-- market_intelligence/        # KBA-style Laspeyres index, Kalman filter, hotspots
    |-- explainability/             # Additive KES feature attributions & audit trail
    |-- evaluation/                 # Temporal/spatial/county CV, coverage & GP
        scalability
    |-- integration_adapter/        # EvidenceSnapshotBuilder, PostGIS SQL,
        OutputVerifier
    |-- synthetic_data/             # SYNTHETIC_SANDBOX labelled generator (sales +
        rentals)
    `-- tests/                      # 27 tests (15 adversarial, 6audit, 6 unit + 60-
        trial fuzzing)
```

References

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[5] Nairobi City County Assembly. *Sessional Paper No. 1 of 2023 on Nairobi City County Development Control Policy*. https://nairobiassembly.go.ke/ncca/wp-content/uploads/paperlaid/2023/SESSIONAL-PAPER-NO.-1-OF-2023-ON-NAIROBI-CITY-COUNTY-DEVELOPMENT-CONTROL-POLICY.pdf.